

Society Lab

Computational social science for societies of AI agents.

Contributors: **Atharva Pandey** and **Gautam Jajoo**. The current research uses the AI Village dataset from AI Digest. Gautam's Observatory reference informed the adapted, explicitly unvalidated behavior rubrics and source-reference design.

Agent teams can report progress while a collaborator cannot open the work. Is the problem a copied URL, access rights, browser context, or coordination around the wrong document? Society Lab makes that ambiguity researchable: connect real traces, inspect evidence, compare explanations, build an explicit world, intervene, execute real agents, and replay the result.

Start with real AI Village evidence

The current source contains 2,000 retained September messages: 1,983 agent posts and 17 user posts. These are reported events, not authenticated historical browser or Google-account state. An actual source-search tool scans the exact retained dataset, reports literal match counts and centers excerpts on their original text coordinates. It found reported 404 and Page Not Found incidents; introductory samples cannot establish absence.

A real investigator and independent skeptic examined the access pattern. The proposed distinct social mechanism was rejected: ordinary reference transcription, browser/session friction and routine shared-document work remained sufficient explanations. Novelty and historical causal support are not established. The lab preserves that negative judgment rather than promoting an attractive story.

A narrower, executable question

Can two agents repair a broken link so the teammate actually opens the original? This new experiment holds content, viewer access and starting authenticated profiles fixed. One auditor reference starts broken. A document owner and teammate can search by title, navigate an exact reference, inspect and message through real local tools. Both arms receive the same clear common goal: the auditor must open the current correct original. Copying, sharing and four-document completion are outside this task.

A real research builder and causal reviewer check the legal actions, hypothesis and source-bound fidelity contract. The world explicitly approximates browser/reference mechanics and omits original Google services, hidden history and historical model policies. Source observations motivate the test; they do not establish simulator equivalence.

The intervention compares an equal-word-count canonical-reference-check note with a neutral note. Two matched pairs share generated starting worlds; their fresh teams are isolated and arm order is randomized. Within-team interactions are allowed. Visible context is placed in actual subject requests; this does not prove private thought editing or consumption.

What the real run found

Four fresh two-role teams made 64 real LLM decisions. All four solved the registered task: 2/2 under each note. Each auditor actually opened the current original. The paired effect was zero, with a saved 95% bounded-pair Hoeffding interval from -100 to +100 percentage points. This demonstrates a usable legal repair task and functioning end-to-end workflow; it does not demonstrate note benefit or equivalence.

Fresh deterministic replay and exact fact checks passed. All 64 unique provider responses were independently matched to recorded actions. Only one team had a separate auditor content-inspection receipt; the primary checks executed opening, correct current content and authorization, not reading or

understanding. The previous broader task's floor result motivated task/goal debugging; its outcome was not relabeled as success or pooled with this new protocol.

The product and startup direction

A persistent AI guide uses actual tools to connect sources, retrieve evidence and carry out explicitly authorized research. Projects, chats, exact saved versions, editable copies and cross-chat reuse preserve provenance. Animated replay displays real posts or executed actions; addressed messages are not assumed to have been read. Native Matplotlib plots show matched outcomes, uncertainty and ordinal tool progression without invented durations.

Graph, temporal-path, spectral and Hodge instruments support routing questions. Algebra is a measurement layer, not a causal network or behavioral diagnosis. A behavior/theory library preserves rival explanations, falsifiers and rejected candidates. Optional Laya/Jev interfaces provide a scale path; local Laya was memory-blocked, so no accuracy, calibration or inference-speed claim is made.

The startup idea is a laboratory for agent societies: source-grounded observability feeds a fidelity-aware wind tunnel, and proposed explanations stay linked to tests that could refute them.

Using the tool

Society Lab is a branching workspace, not a compulsory wizard. A project groups sources, questions and artifacts. Each working chat retains conversation, selected context and drafts. You can inspect a result, revisit its source, branch a rival hypothesis or switch chats while a task runs. Background jobs keep the origin chat captured at launch.

Start with **logs only** or **logs plus an environment**. Logs alone support observation and hypotheses; experiments need a separately reviewed world. A supplied task, files or service contract can inform an adapter, but uploaded code is not automatically executed. The builder must expose scoped observations, reset, actions and state transitions, communication rules, costs and an independent outcome oracle. Missing original tools may use disclosed proxy semantics. Neither schema validity nor visual fidelity proves historical equivalence.

Connect saved chat or actual `societylab.events.v1` telemetry. The event protocol records explicit actors, task transitions, recipients, tool calls/returns, artifacts and context delivery. Stable IDs permit exact retries and immutable updates; missing fields remain unknown. Producer timestamps, success reports and addressed recipients are not independently authenticated historical outcomes or readership. The current limit is 2,000 accumulated events and 1 MiB per run.

The source brief begins with cheap recorded-field and lexical screening. Real investigator and skeptic agents then retrieve exact messages, context, ordinary examples and counterexamples. The six research responsibilities are discovery, skeptic, environment builder, causal methodologist, evaluator and theory curator. Their prompts, tools and skill files define bounded harnesses. Responses provides structured outputs and function calls; the optional Codex research adapter reads an isolated evidence packet. Fresh experimental subjects currently use Responses.

Collaborate through the AI guide

The guide is always available alongside the workspace. It can explain saved context, navigate, read another chat and use real tools to connect sources, investigate, save a plan, build/review a simulator, run subjects, manage projects and drafts, or save and measure a rubric. General discussion does not authorize a paid experiment. Explicit requests such as "Build this simulator; do not run subjects yet" keep preparation and execution separate.

Type **@** to select an exact saved item or working chat. A selected chip carries identity; a title typed in prose does not. Reading another chat borrows a pinned discussion snapshot while keeping the current chat.

Reusing an artifact links its same exact version. Making a copy creates an editable draft with origin and excludes empirical outcomes and verification. Forking a discussion creates a working branch, not another experimental observation. Frozen sources, protocols and results are never edited through a draft.

The activity feed shows actual started, completed, refused and failed operations, with their returned references. Working summaries describe operations, not hidden model reasoning. Successful tool receipts, source cards and results remain in the chat after reload. If an operation is interrupted or uncertain, it is not automatically rerun. An apparent claim of completion should be checked against its receipt.

Useful requests include "Find ordinary explanations for these access reports," "Save a narrow source-linked hypothesis," "Show the actual world tools and what they omit," "Run this reviewed world," and "Reuse @that result here without copying it." The current demonstration's actual plan, world and result are exact linked artifacts rather than independently selected items with similar names.

Review an experiment before execution

A plan states its hypothesis and rivals, experimental unit, subject model, conditions, information boundaries, intervention boundary, action budget, reset and analysis. The builder maps it to executable capabilities; the causal reviewer may require changes or decline. The saved protocol freezes world/code, notes, model settings and assignment before outcomes. Underpowered exploratory work is labeled as such, rather than presented as confirmation.

The new document-recovery world activates only owner and auditor. Content, viewer ACL and initial authenticated profiles are fixed, and a single copied reference is broken. Ten tools support exact navigation, title search, profile switching, clipboard, inspection, direct messaging, reporting and waiting. Both arms get the same tools, clear common goal and opportunity budget. Private context adds visible text to the subject request; it does not edit hidden thoughts or prove consumption.

Whole teams are randomized, with isolated matched worlds and randomized arm order. Within-team communication is part of the policy package. Individual agent decisions are not independent statistical replicates. Invalid choices consume opportunities; infrastructure failures retain assignments and partial traces, with no fabricated completion estimate. A successful local proxy test does not identify which historical cause produced the source incident.

The developer framework also includes artifact verification, correlated-source diffusion, complementary-information tasks, exclusive-resource work, observable-trigger context timing and correction routing. Each has different capabilities and estimands. These are optional reviewed test families, not a premature taxonomy of agent behavior or interchangeable substitutes for a source question. More realistic supplied environments require explicit adapters and fidelity review.

The behavior library and measurement layer

The library retains candidates, skeptical rejections, source evidence, ordinary alternatives, falsifiers, review attempts, linked experiments and bounded theories. Evidence status is separate from novelty and causal support. A pilot-tested or replication-tested record does not automatically establish a mechanism or generalization. Supporting and conflicting outcomes remain together; rejected hypotheses are useful results.

Eight proposed starter rubrics cover contributions, qualified information, handoffs, commitments, advice, correction, returned help and shared-resource rules. Researchers can save prompts with positive, negative, unknown and non-example rules, then screen an exact dataset with an exact rubric version. These are original compact adaptations of an Observatory reference, not a validated social scale.

Laya is the configured local prompt-defined text-screen interface. Failure, missing runtime, RAM limits and abstention stay unknown; no regex fallback supplies a negative label. On this laptop the RAM guard blocked inference, so no local accuracy or speed claim is made. Optional Jev-like/provider interfaces do not establish verifier performance. A message-level screen cannot supply a verified opportunity, recipient exposure, task

completion or motivation. A separate review workflow freezes a seeded binary-question sample and records yes/no/uncertain judgments with declared reviewer provenance; agent-assisted labels are not human ground truth.

Read results, replay and graph mathematics

Result views distinguish team counts from rates, primary outcomes from process measures, and checked numerical facts from unverified explanations. The current four-team success result has a zero contrast and a wide interval; success in both arms is not evidence of equivalence. Post-treatment centrality, messages or information coverage are not automatically identified causal mediators.

Replay follows actual recorded messages or executed subject decisions. Prefix state shows only events reached by the playhead. Local/private records, direct sends, room posts, declared broadcasts and unknown audiences have distinct lanes. Declared addressing is not reading; a retained rationale is not hidden thought. Safe embedded reports are rendered from exact saved results, not arbitrary model HTML.

Communication graphs describe mentions or declared addresses under an explicit instrument. Event neighborhoods follow unique typed source references; artifact maps follow saved research provenance. Temporal paths, spectral/Laplacian signals and Hodge components are useful exploratory measurements, not evidence of influence or a diagnosed social role. Extraction, node universe, timing, missingness and source scope travel with each result.

Independent replay reconstructs saved requests, tools and outcome oracles under pinned code. Finite checks validate declared counts and effects, not every free-prose explanation, hidden provider consumption or historical Google state. Replaying a result uses no new subjects and is not replication. Changed questions, worlds or backends need new registrations rather than retrospective edits.

Set up and continue

Run from a complete source checkout with Python 3.11+, retaining web, prompts, skills and research. NumPy enables the graph mathematics; Matplotlib enables native plots. Supply `OPENAI_API_KEY` through the environment or `.secrets/openai-api-key.txt`, not a chat or committed file. A clean checkout has no private runtime sources.

```
python -m pip install numpy matplotlib
python scripts/configure-guide.py --research-calls 400 --guide-calls 100
python -m swarm_lab.cli --max-calls 400 serve --port 8765
```

Open `http://127.0.0.1:8765`. The two research-cap values must match. These are explicit example allowances; the persistent call ledger is not reset by configuration or restart. The local API uses bounded strict JSON, exact versions/hashes, session/CSRF checks and captured chat origins. It is a local research product, not a publicly exposed multi-user deployment. If a source, tool or budget is unavailable, the failure remains visible rather than silently changing the method.

[Full user guide](#) · [Demonstration](#) · [Exact results and limitations](#) · [New solved-task result](#).